Quiz — 1.3.3 Networks

Name: ____ Date: _ ***Mark:*** ___ / 25

OCR H446 · Component 01 · 1.3.3

Section A — Multiple choice (8 marks)

1. Which best describes a LAN? A. A network spanning cities or countries using third-party infrastructure B. A network over a small geographical area where the organisation owns the hardware C. A network with no physical hardware D. The largest WAN in the world

Your answer: ☐

2. In a peer-to-peer network: A. A central server provides all services B. All devices are equal and can act as client and server C. There is always a single point of failure D. Security is centrally managed

Your answer: ☐

3. A disadvantage of a star topology is that: A. A single cable fault brings down every device B. The central switch/hub is a single point of failure C. It cannot use a switch D. It requires every node to connect to every other node

Your answer: ☐

4. In packet switching, packets are: A. Sent along one dedicated reserved path B. Routed independently and reassembled at the destination C. Never reassembled D. Only used for telephone calls

Your answer: ☐

5. Which protocol is used to *send* email? A. POP3 B. IMAP C. SMTP D. FTP

Your answer: ☐

6. The DNS is responsible for: A. Encrypting web traffic B. Translating domain names into IP addresses C. Splitting data into packets D. Assigning MAC addresses

Your answer: ☐

7. Which device connects two different networks and routes using IP addresses? A. Switch B. Router C. NIC D. Wireless access point

Your answer: ☐

8. At which TCP/IP layer are source and destination IP addresses added? A. Application B. Transport C. Network/Internet D. Link/Data link

Your answer: ☐

Section B — Short answer (12 marks)

9. State two advantages of a client-server network over a peer-to-peer network. **[2]**

10. TCP/IP layer matching. Match each item (A–D) to the correct layer by writing the layer name next to it. **[4]**

A. Adds the MAC address / handles the physical connection → ________

B. Adds source and destination IP addresses and routes packets → ________

C. Splits data into packets, manages ports and end-to-end delivery → ________

D. Provides services for user applications (e.g. HTTP, SMTP) → ________

11. Explain the difference between a switch and a router. **[2]**

12. Describe what a proxy server does and give one benefit of using one. **[2]**

13. Explain one reason packet switching is more resilient than circuit switching. **[2]**

Section C — Exam-style question (5 marks)

14. A user types a web address into their browser and requests a secure web page over the internet.

Describe what happens, referring to: the role of DNS, how the data is broken up and routed across the network, and one piece of hardware involved. **[5]**

Answer key

Section A (8 marks, 1 each)

1. **B** — LAN = small area, single site, organisation-owned hardware.
2. **B** — In P2P all peers are equal, acting as both client and server.
3. **B** — The central switch/hub is the single point of failure in a star.
4. **B** — Packets are routed independently and reassembled at the destination.
5. **C** — SMTP sends mail; POP3/IMAP retrieve it.
6. **B** — DNS translates domain names to IP addresses.
7. **B** — A router connects different networks and routes using IP.
8. **C** — The Network/Internet layer adds IP addressing and routes packets.

Section B (12 marks)

9. **[2]** Any two for 1 mark each: centralised security/backups; easier to manage/update centrally; scales well to many users; central data storage/control.

10. **[4] (1 mark each)** - A → **Link / Data link** layer - B → **Network / Internet** layer - C → **Transport** layer - D → **Application** layer

11. **[2]** 1 mark: a switch connects devices *within* a single LAN and forwards frames using MAC addresses. 1 mark: a router connects *different* networks (e.g. LAN to internet) and forwards packets using IP addresses.

12. **[2]** 1 mark: a proxy acts as an intermediary between client and server, forwarding requests/responses. 1 mark for one benefit: hides the user's IP address, caches pages to speed up access, or filters/logs/monitors traffic.

13. **[2]** 1 mark: packets can be routed independently along different routes. 1 mark: if one route/node fails, packets can be re-routed around it (no single dedicated path to break), so the communication still gets through.

Section C (5 marks — award up to 5 from)

14.
    - DNS translates the typed domain name into the server's IP address (querying DNS servers, results may be cached). **[1]**
    - The browser (client) sends a request to that IP address; HTTPS is used so the data is encrypted in transit. **[1]**
    - The data is split into packets at the transport layer, each given a header (e.g. sequence number, ports). **[1]**
    - Packets are given source/destination IP addresses and routed independently across the network, then reassembled in order at the destination. **[1]**
    - A router (and/or switch/NIC) forwards the packets between networks toward the destination using IP addresses. **[1]**

Total: 8 + 12 + 5 = 25 marks